

Raghav Malik

PHD CANDIDATE · COMPUTER ENGINEERING

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Education

Purdue University

PHD COMPUTER ENGINEERING

• Advisor: Dr. Milind Kulkarni

BS COMPUTER ENGINEERING

West Lafayette

August 2019 - present

August 2016 - May 2019

Professional Experience

Research Assistant, Parallelism, Languages, and Compilers Lab

2019-2024

- Developed a novel abstraction for expressing homomorphic circuits based on programmable bootstrapping
- Designed a protocol for evaluating computations with ciphertext-dependent branches
- Extended general vectorization techniques to apply to encrypted arithmetic circuits
- Developed an algorithm for efficiently vectorizing private decision tree inference
- Contributed to the design of a metaprogramming language for writing cryptographic applications

Research Intern, Microsoft Research:

2022

- Developed scheduling optimizations for the MSCCL GPU compiler
- Designed an event-based simulation engine for profiling MSCCL communication collectives

Software Intern, Amazon Robotics:

2021

- Designed a DSL for expressing complex data validation rules

Software Intern, AWS Kinesis:

2019-2020

- Developed features for Kinesis Firehose for monitoring and streaming to S3

Publications

Malik, Raghav, Vedant Paranjape, Milind Kulkarni. 2025. COATL: Circuit Optimization using Arithmetic Table Lookup. <https://raghav198.github.io/assets/papers/coatl.pdf> [Accepted to PLDI 2025]

Malik, Raghav, Kabir Sheth, Milind Kulkarni. 2023. Coyote: A Compiler for Vectorizing Encrypted Arithmetic Circuits. Proceedings of the 28th ACM International Conference on Architectural Support for Programming Languages and Operating Systems, Volume 3: 118-133.

Malik, Raghav, Vidush Singhal, Benjamin Gottfried, Milind Kulkarni. 2021. Vectorized Secure Evaluation of Decision Forests. Proceedings of the 42nd ACM SIGPLAN International Conference on Programming Language Design and Implementation: 1049-1063.

Bao, Yuyan, Kirshanthan Sundarajah, **Raghav Malik**, Qianchuan Ye, et. al. 2021. HACCLE: Metaprogramming for Secure Multi-Party Computation. Proceedings of the 20th ACM SIGPLAN International Conference on Generative Programming: Concepts and Experiences: 130-143.

PREPRINTS

Malik, Raghav, Milind Kulkarni. 2024. COIL: Compiling Homomorphic Circuits with Control Flow. <https://raghav198.github.io/assets/papers/coil.pdf>

Sreekanth, Sreevickrant, Dulani Wijayarathne, **Raghav Malik**, Milind Kulkarni. 2024. Biscotti: An Approach to Parallel Scheduling for Vectorized Encrypted Arithmetic Circuits. <https://raghav198.github.io/assets/papers/biscotti.pdf>

Presentations

* *presenting author*; + *mentored undergraduate*

INVITED TALKS

Fall 2024. *Compiling Homomorphic Circuits with Control Flow*. Poster: Midwestern PL Summit 2024

Fall 2023. *Managing Ownership and Lifetimes*. Guest lecture, Object Oriented Programming in C++, Purdue University

Fall 2023. *Coyote: A Compiler for Vectorizing Encrypted Arithmetic Circuits*. Invited talk: Midwestern PL Summit 2023

Spring 2023. *Coyote: A Compiler for Vectorizing Encrypted Arithmetic Circuits*. Invited talk: Galois, Inc.

Fall 2022. *Vectorized Secure Evaluation of Decision Forests*. Invited talk: Cornell University.

CONTRIBUTED PRESENTATIONS

Sreekanth, Vickranth⁺, Dulani Wijayarathne⁺, **Raghav Malik**, Milind Kulkarni. 2023. Parallel Encrypted Arithmetic Vector Scheduling for Optimized FHE Computations. Departmental seminar: Purdue Programming Languages Group.

Sreekanth, Vickranth⁺, Dulani Wijayarathne⁺, **Raghav Malik**, Milind Kulkarni. 2023. An Optimization for Vectorizing Encrypted Arithmetic Circuits. Poster: Summer Undergraduate Research Symposium, Purdue University.

Bao, Yuyan, Kirshanthan Sundarajah^{*}, **Raghav Malik**, Qianchuan Ye, et. al. 2021. HACCLE: Metaprogramming for Secure Multiparty Computation. GPCE Conference Talk.

Projects

CATEGORIAL NERVE IN LEAN

Used Lean to define and formalize explicit constructions of the nerve of a category and the homotopy realization of a simplicial set, and proved an adjunction between these functors.

PROOFMOUSE

A lightweight proof assistant for verifying Gentzen-style propositional and predicate logic proofs. ProofMouse is currently being used as a teaching aid to help discrete mathematics students learn logic and how to write proofs.

Teaching Experience

Fall 2024	Advanced C Programming , Teaching Assistant <i>Designed problem sets and exams, delivered some lectures, held office hours</i>
Fall 2024	Object Oriented Programming in C++ , Teaching Assistant <i>Designed and delivered some lectures, wrote exams, managed undergraduate TAs</i>
Summer 2024	Data Structures and Algorithms , Instructor <i>Designed an 8-week course, designed and delivered lectures, problem sets, and exams.</i>
Spring 2024	Object Oriented Programming in C++ , Teaching Assistant <i>Held office hours, managed undergraduate TAs, designed and delivered some lectures</i>
Fall 2023	Object Oriented Programming in C++ , Teaching Assistant <i>Held office hours, developed autograding framework, delivered guest lecture</i>
Spring 2023	Discrete Mathematics , Teaching Assistant <i>Held office hours, developed a proof assistant for automatically verifying Gentzen-style proofs</i>

Mentoring

2022-2024	Dulani Wijayarathne , Summer Undergraduate Research Student
2022-2024	Vickranth Sreekanth , Summer Undergraduate Research Student
2021-2022	Kabir Sheth , Undergraduate Research Assistant
2019-2021	Vidush Singhal , Undergraduate Research Assistant
2019-2020	Benjamin Gottfried , Undergraduate Research Assistant

Skills

TECHNICAL SKILLS

C++, Python, OCaml, Haskell, Linux, Git, object-oriented programming, functional programming

RESEARCH SKILLS

Compiler optimizations, vectorization, cryptography, homomorphic encryption, multiparty computation, program synthesis

SOFT SKILLS

Mentoring undergraduates, designing and teaching courses, writing grants, collaborating on research

Professional Development

SERVICE

2024-2025 **Association of Computing and Machinery, PLDI**, Organizing Committee, Student Volunteer
Chair

2022-2023 **Association of Computing and Machinery, PLDI**, Student Volunteer

PEER REVIEW

Program Committee, ARRAY 2025

Artifact Evaluation Committee, PPOPP 2024